

Heat

Multiple choice questions

1. What are all substances made up of?

- a) Cells
- b) Atoms and molecules
- c) Rocks
- d) Liquids

Answer: b) Atoms and molecules

2. Why do particles possess heat energy?

- a) Because they are heavy
- b) Because they are colorful
- c) Because they are always moving
- d) Because they are large

Answer: c) Because they are always moving

3. What is heat energy?

- a) Energy of sound
- b) Energy due to motion of particles
- c) Energy of light
- d) Energy of electricity only

Answer: b) Energy due to motion of particles

4. Heat is a form of energy that is transferred due to difference in what?

- a) Weight
- b) Shape
- c) Temperature
- d) Size

Answer: c) Temperature

5. Heat usually flows from a _____ object to a _____ object.

- a) Cold, hot
- b) Small, big
- c) Hot, cold
- d) Light, heavy

Answer: c) Hot, cold

6. Which of the following is an example of heat energy?

- a) Playing football
- b) Ironing clothes
- c) Reading a book
- d) Writing

Answer: b) Ironing clothes

7. Which appliance uses electrical energy to produce heat?

- a) Fan
- b) Television
- c) Iron box
- d) Computer

Answer: c) Iron box

8. Heat is not a matter because it does not have:

- a) Energy
- b) Temperature
- c) Weight
- d) Motion

Answer: c) Weight

9. Which SI unit is used to measure heat energy?

- a) Kelvin
- b) Joule
- c) Meter
- d) Watt

Answer: b) Joule

10. Heat energy can also be measured in:

- a) Newton
- b) Pascal
- c) Calorie
- d) Volt

Answer: c) Calorie

11. Which is the major natural source of heat?

- a) Fire
- b) Electricity
- c) Sun

d) Coal

Answer: c) Sun

12. Heat from the sun reaches the earth mainly in the form of:

a) Sound

b) Water

c) Light

d) Air

Answer: c) Light

13. Heat produced in the sun is due to:

a) Combustion

b) Friction

c) Nuclear fusion

d) Electricity

Answer: c) Nuclear fusion

14. What is combustion?

a) Cooling process

b) Burning of fuel

c) Melting of solids

d) Freezing of liquids

Answer: b) Burning of fuel

15. Which of the following is a fuel?

a) Water

b) Air

c) Wood

d) Sand

Answer: c) Wood

16. Which energy is produced during combustion?

a) Sound energy

b) Light energy only

c) Heat energy

d) Magnetic energy

Answer: c) Heat energy

17. Flow of electrons through a substance is called:

- a) Heat
- b) Temperature
- c) Electric current
- d) Energy

Answer: c) Electric current

18. Materials that allow current to pass through them are called:

- a) Insulators
- b) Conductors
- c) Resistors
- d) Batteries

Answer: b) Conductors

19. Heat produced in electric wires is due to:

- a) Friction
- b) Resistance
- c) Magnetism
- d) Gravity

Answer: b) Resistance

20. Friction converts _____ energy into heat energy.

- a) Light
- b) Electrical
- c) Kinetic
- d) Sound

Answer: c) Kinetic

21. Why do our hands feel warm when rubbed together?

- a) Due to electricity
- b) Due to friction
- c) Due to light
- d) Due to pressure

Answer: b) Due to friction

22. Which instrument is used to measure temperature?

- a) Barometer
- b) Thermometer
- c) Calorimeter

d) Voltmeter

Answer: b) Thermometer

23. Using touch to judge temperature is not correct because:

a) Fingers are small

b) Touch is inaccurate

c) Heat is invisible

d) Temperature is constant

Answer: b) Touch is inaccurate

24. What is temperature?

a) Amount of heat

b) Measure of hotness or coldness

c) Flow of heat

d) Total energy

Answer: b) Measure of hotness or coldness

25. SI unit of temperature is:

a) Celsius

b) Fahrenheit

c) Kelvin

d) Joule

Answer: c) Kelvin

26. Which unit is also called centigrade?

a) Kelvin

b) Fahrenheit

c) Celsius

d) Joule

Answer: c) Celsius

27. Which temperature scale is mostly used in science?

a) Celsius

b) Fahrenheit

c) Kelvin

d) Rankine

Answer: c) Kelvin

28. Freezing point of water in Celsius scale is:

a) 100°C

b) 32°C

- c) 0°C
- d) 273°C

Answer: c) 0°C

29. Boiling point of water in Celsius scale is:

- a) 0°C
- b) 50°C
- c) 100°C
- d) 212°C

Answer: c) 100°C

30. Freezing point of water in Fahrenheit scale is:

- a) 0°F
- b) 32°F
- c) 100°F
- d) 212°F

Answer: b) 32°F

31. Boiling point of water in Fahrenheit scale is:

- a) 100°F
- b) 180°F
- c) 200°F
- d) 212°F

Answer: d) 212°F

32. Which temperature scale has absolute zero?

- a) Celsius
- b) Fahrenheit
- c) Kelvin
- d) None

Answer: c) Kelvin

33. Absolute zero is equal to:

- a) 0°C
- b) 0°F
- c) 0 K
- d) 100 K

Answer: c) 0 K

34. When water boils, its temperature:

- a) Decreases

- b) Increases continuously
- c) Remains constant
- d) Becomes zero

Answer: c) Remains constant

35. The temperature at which water boils is called:

- a) Freezing point
- b) Melting point
- c) Boiling point
- d) Evaporation point

Answer: c) Boiling point

36. Which instrument measures heat energy?

- a) Thermometer
- b) Calorimeter
- c) Ammeter
- d) Barometer

Answer: b) Calorimeter

37. Heat depends on temperature and:

- a) Colour
- b) Shape
- c) Number of particles
- d) Size only

Answer: c) Number of particles

38. Temperature depends on:

- a) Total kinetic energy
- b) Average kinetic energy
- c) Number of molecules
- d) Volume

Answer: b) Average kinetic energy

39. Symbol used to represent heat is:

- a) T
- b) Q
- c) H
- d) K

Answer: b) Q

40. Symbol used to represent temperature is:

- a) Q
- b) J
- c) T
- d) C

Answer: c) T

41. Which of the following can do work?

- a) Temperature
- b) Heat
- c) Thermometer
- d) Celsius

Answer: b) Heat

42. When heat is added to an object, what happens first?

- a) Colour change
- b) Rise in temperature
- c) Loss of energy
- d) Freezing

Answer: b) Rise in temperature

43. Heating water above 100°C results in:

- a) Ice formation
- b) Vapour formation
- c) Solidification
- d) Cooling

Answer: b) Vapour formation

44. Change from liquid to gas is called:

- a) Condensation
- b) Freezing
- c) Evaporation
- d) Melting

Answer: c) Evaporation

45. Flow of heat stops when two objects reach:

- a) Same mass
- b) Same shape
- c) Thermal equilibrium

d) Same size

Answer: c) Thermal equilibrium

46. Two systems having same temperature are said to be in:

a) Thermal contact

b) Thermal expansion

c) Thermal equilibrium

d) Heat transfer

Answer: c) Thermal equilibrium

47. Heat transfer occurs only when there is a difference in:

a) Mass

b) Shape

c) Temperature

d) Colour

Answer: c) Temperature

48. In a hot coffee and hand example, heat flows from:

a) Hand to coffee

b) Coffee to hand

c) Both ways

d) Neither

Answer: b) Coffee to hand

49. Expansion of a substance on heating is called:

a) Contraction

b) Compression

c) Thermal expansion

d) Evaporation

Answer: c) Thermal expansion

50. On cooling, substances generally:

a) Expand

b) Contract

c) Melt

d) Boil

Answer: b) Contract

51. Which material expands more on heating?

- a) Glass
- b) Plastic
- c) Steel
- d) Rubber

Answer: c) Steel

52. Why does a metal lid open easily after hot water is poured?

- a) Glass expands more
- b) Lid expands more
- c) Glass melts
- d) Water cools it

Answer: b) Lid expands more

53. Iron rim is heated before fitting on wooden wheel to:

- a) Contract it
- b) Melt it
- c) Expand it
- d) Cool it

Answer: c) Expand it

54. After cooling, the iron rim on the wheel:

- a) Loosens
- b) Falls off
- c) Contracts and tightens
- d) Melts

Answer: c) Contracts and tightens

55. Riveting works on the principle of:

- a) Cooling
- b) Friction
- c) Thermal expansion
- d) Electricity

Answer: c) Thermal expansion

56. Hot rivets after cooling:

- a) Expand
- b) Shrink
- c) Break

d) Melt

Answer: b) Shrink

57. Why are gaps left between railway tracks?

a) For decoration

b) For rainwater

c) For thermal expansion

d) For strength

Answer: c) For thermal expansion

58. If gaps are not provided in rails, it may cause:

a) Cooling

b) Expansion

c) Bending of tracks

d) Painting

Answer: c) Bending of tracks

59. Expansion in length of a solid is called:

a) Areal expansion

b) Cubical expansion

c) Linear expansion

d) Random expansion

Answer: c) Linear expansion

60. Expansion in area of a solid is called:

a) Linear

b) Areal

c) Cubical

d) Volumetric

Answer: b) Areal

61. Expansion in volume of a solid is called:

a) Linear

b) Areal

c) Cubical

d) Superficial

Answer: c) Cubical

62. Coefficient of linear expansion is represented by:

a) β

b) γ

c) α

d) θ

Answer: c) α

63. Coefficient of areal expansion is represented by:

a) α

b) β

c) γ

d) δ

Answer: b) β

64. Coefficient of volumetric expansion is represented by:

a) α

b) β

c) γ

d) λ

Answer: c) γ

65. Which expansion occurs only in solids?

a) Linear

b) Areal

c) Cubical

d) All of these

Answer: d) All of these

66. Heat transfer requires:

a) Motion

b) Temperature difference

c) Light

d) Electricity

Answer: b) Temperature difference

67. When two bodies touch and exchange heat, they are in:

a) Mechanical contact

b) Thermal contact

c) Chemical contact

d) Electrical contact

Answer: b) Thermal contact

68. The direction of heat flow is determined by:

a) Mass

- b) Volume
- c) Temperature
- d) Density

Answer: c) Temperature

69. Heat energy increases the _____ of particles.

- a) Colour
- b) Weight
- c) Speed
- d) Shape

Answer: c) Speed

70. Average kinetic energy of particles represents:

- a) Heat
- b) Temperature
- c) Mass
- d) Volume

Answer: b) Temperature

71. Total kinetic energy of particles represents:

- a) Temperature
- b) Heat
- c) Density
- d) Pressure

Answer: b) Heat

72. Which of the following is NOT a source of heat?

- a) Sun
- b) Electricity
- c) Friction
- d) Moon

Answer: d) Moon

73. Which process converts work into heat?

- a) Evaporation
- b) Friction
- c) Condensation
- d) Freezing

Answer: b) Friction

74. When electrons face resistance, electrical energy converts into:

- a) Light
- b) Sound
- c) Heat
- d) Motion

Answer: c) Heat

75. Temperature does NOT depend on:

- a) Particle motion
- b) Average kinetic energy
- c) Number of molecules
- d) Speed of particles

Answer: c) Number of molecules

76. Heat is measured using:

- a) Thermometer
- b) Calorimeter
- c) Ammeter
- d) Voltmeter

Answer: b) Calorimeter

77. Temperature is measured using:

- a) Calorimeter
- b) Barometer
- c) Thermometer
- d) Hygrometer

Answer: c) Thermometer

78. Heat energy is a form of:

- a) Matter
- b) Force
- c) Energy
- d) Substance

Answer: c) Energy

79. Heat has no:

- a) Energy
- b) Temperature
- c) Weight

d) Motion

Answer: c) Weight

80. Which unit is NOT used for temperature?

a) Kelvin

b) Celsius

c) Fahrenheit

d) Joule

Answer: d) Joule

81. Which unit is NOT used for heat?

a) Joule

b) Calorie

c) Kelvin

d) All are correct

Answer: c) Kelvin

82. Heating a substance mainly increases:

a) Mass

b) Molecular motion

c) Colour

d) Shape

Answer: b) Molecular motion

83. Cooling a substance decreases:

a) Molecular motion

b) Mass

c) Volume always

d) Weight

Answer: a) Molecular motion

84. At boiling point, water changes into:

a) Ice

b) Liquid

c) Vapour

d) Solid

Answer: c) Vapour

85. Temperature remains constant during:

a) Heating always

b) Cooling always

- c) Change of state
- d) Expansion

Answer: c) Change of state

86. Which scale is commonly used in India?

- a) Fahrenheit
- b) Kelvin
- c) Celsius
- d) Rankine

Answer: c) Celsius

87. Heat energy always flows:

- a) Randomly
- b) From cold to hot
- c) From hot to cold
- d) In circles

Answer: c) From hot to cold

88. Expansion depends on:

- a) Colour of object
- b) Temperature change
- c) Shape only
- d) Density only

Answer: b) Temperature change

89. Which expands least on heating?

- a) Metal
- b) Gas
- c) Liquid
- d) Glass

Answer: d) Glass

90. Linear expansion deals with change in:

- a) Area
- b) Volume
- c) Length
- d) Mass

Answer: c) Length

91. Areal expansion deals with change in:

- a) Length
- b) Area
- c) Volume
- d) Weight

Answer: b) Area

92. Cubical expansion deals with change in:

- a) Length
- b) Area
- c) Volume
- d) Temperature

Answer: c) Volume

93. Heat from sun does not remain near sun because it:

- a) Is heavy
- b) Spreads into space
- c) Freezes
- d) Stops

Answer: b) Spreads into space

94. Which is an artificial source of heat?

- a) Sun
- b) Firewood
- c) Nuclear fusion
- d) Stars

Answer: b) Firewood

95. Combustion needs oxygen because it acts as:

- a) Fuel
- b) Catalyst
- c) Oxidizer
- d) Product

Answer: c) Oxidizer

96. When rails expand in summer, gaps help to:

- a) Break tracks
- b) Prevent bending
- c) Increase heat

d) Cool rails

Answer: b) Prevent bending

97. Thermal equilibrium means:

a) Same heat

b) Same mass

c) Same temperature

d) Same size

Answer: c) Same temperature

98. Which process shows thermal expansion clearly?

a) Melting ice

b) Opening tight metal lid

c) Cooling water

d) Freezing milk

Answer: b) Opening tight metal lid

99. Heat energy of an object depends on:

a) Temperature only

b) Size only

c) Temperature and number of molecules

d) Colour

Answer: c) Temperature and number of molecules

100. Temperature mainly indicates:

a) Total energy

b) Average energy of particles

c) Mass of object

d) Volume of object

Answer: b) Average energy of particles